Introduction to Neural Networks

Artificial Neural Network (ANN)

- Artificial Neural Network (ANN) is a system based on the operation of biological neural networks
- Why would be necessary the implementation of artificial neural networks?
  - Although computing these days is truly advanced, there are certain tasks that a program made for a common microprocessor is unable to perform; even so a software implementation of a neural network can be made

Artificial Neural Network (ANN) Contd

- **Advantages:**
  - It perform tasks that a linear program can not
  - When an element of the neural network fails, it can continue without any problem by their parallel nature
  - A neural network learns and does not need to be reprogrammed
  - It can be implemented in any application
  - It can be implemented without any problem

- **Disadvantages:**
  - The neural network needs training to operate
  - The architecture of a neural network is different from the architecture of microprocessors therefore needs to be emulated
  - Requires high processing time for large neural networks

Artificial Neural Network (ANN) Contd

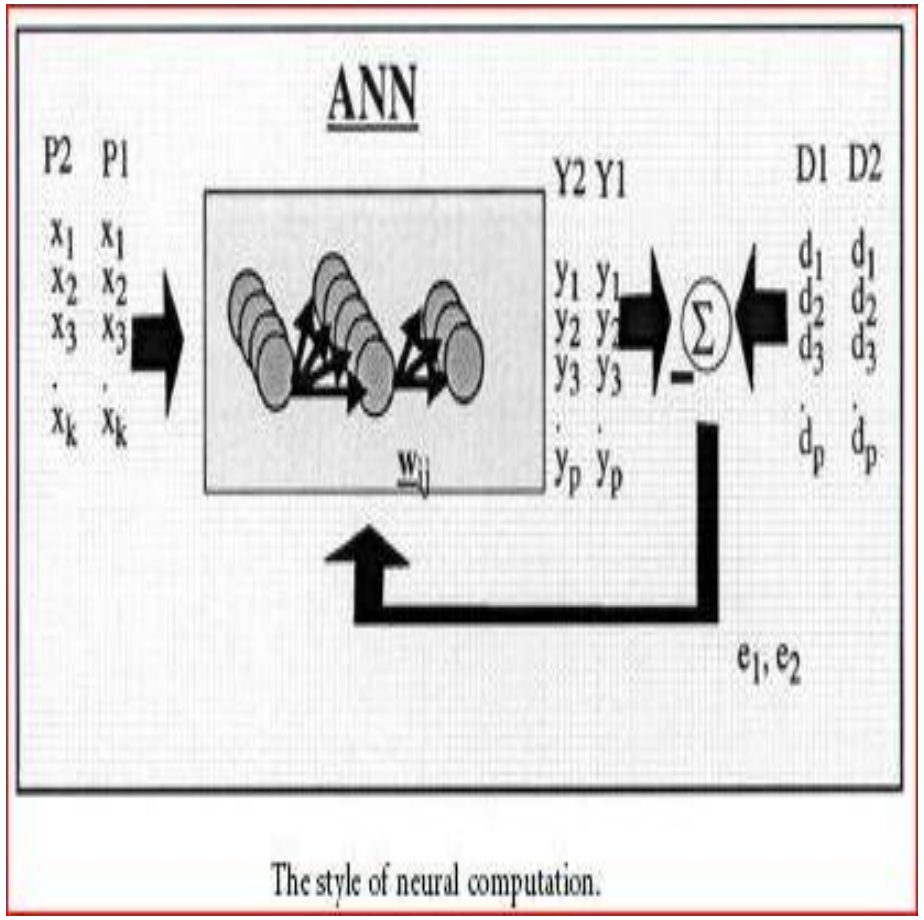
- Artificial neural networks (ANNs) are among the newest signal-processing technologies in the engineer's toolbox
- The field is highly interdisciplinary, but our approach will restrict the view to the engineering perspective
- In engineering, neural networks serve two important functions:
  - as pattern classifiers and
  - as nonlinear adaptive filters
- An ANN is an adaptive, most often nonlinear system that learns to perform a function (an input/output map) from data
  - Adaptive means that the system parameters are changed during operation, normally called the **training phase**

Artificial Neural Network (ANN) Contd

- After the training phase the ANN parameters are fixed and the system is deployed to solve the problem at hand (**the testing phase**)
- The ANN is built with a systematic step-by-step procedure to optimize a performance criterion or to follow some implicit internal constraint, which is commonly referred to as the **learning rule**
- The input/output training data are fundamental in neural network technology
  - because they convey the necessary information to "discover" the optimal operating point
- The nonlinear nature of the neural network processing elements (PEs) provides the system with lots of flexibility to achieve practically any desired input/output map, i.e., some ANNs are universal mappers

Artificial Neural Network (ANN) Contd

- An input is presented to the neural network and
- a corresponding desired or target response set at the output (when this is the case the training is called supervised)
- An error is composed from the difference between the desired response and the system output
- This error information is fed back to the system and adjusts the system parameters in a systematic fashion (the learning rule)
- The process is repeated until the performance is acceptable



The style of neural computation.

Artificial Neural Network (ANN) Contd

- It is clear from this description that the performance hinges heavily on the data
  - If one does not have data that cover a significant portion of the operating conditions or if they are noisy, then **neural network** technology is probably not the right solution
  - On the other hand, if there is plenty of data and the problem is poorly understood to derive an approximate model, then neural network technology is a good choice
- This operating procedure should be contrasted with the traditional engineering design, made of exhaustive subsystem specifications and intercommunication protocols

Artificial Neural Network (ANN) Contd

- In artificial neural networks, the designer chooses:
  - the network topology
  - the performance function
  - the learning rule
  - and the criterion to stop the training phase
  - but the system automatically adjusts the parameters
- So, it is difficult to bring a priori information into the design, and when the system does not work properly it is also hard to incrementally refine the solution
- But ANN-based solutions are extremely efficient in terms of development time and resources
  - and in many difficult problems artificial neural networks provide performance that is difficult to match with other technologies

Artificial Neural Network (ANN) Contd

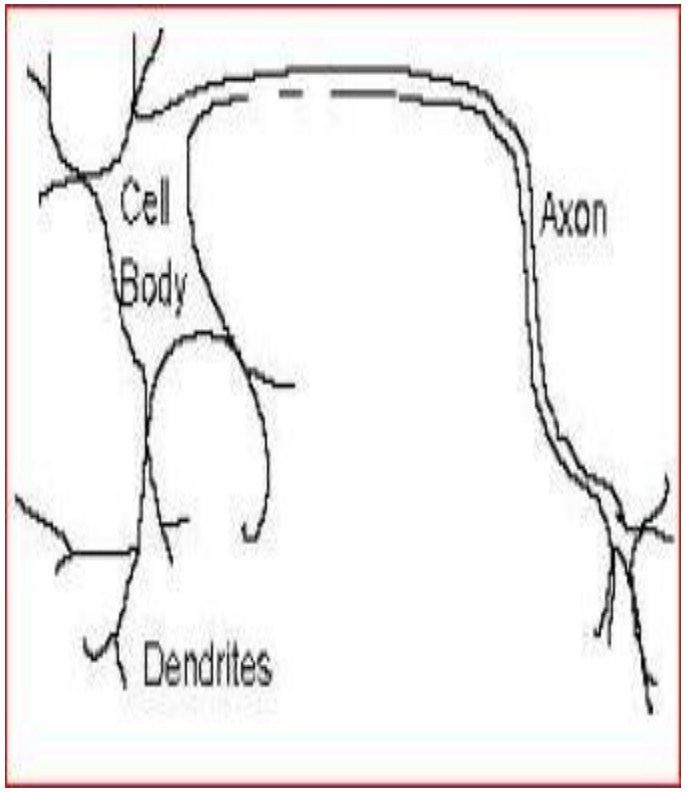
- Denker 10 years ago said that "artificial neural networks are the second best way to implement a solution"
  - motivated by the simplicity of their design and because of their universality
  - only shadowed by the traditional design obtained by studying the physics of the problem
- At present, artificial neural networks are emerging as the technology of choice for many applications, such as
  - pattern recognition
  - Prediction
  - system identification, and control

ANN: The Biological Model

- Artificial neural networks emerged after the introduction of simplified neurons by McCulloch and Pitts in 1943 (McCulloch & Pitts, 1943)
- These neurons were presented as models of biological neurons and as conceptual components for circuits that could perform computational tasks
- The basic model of the neuron is founded upon the functionality of a biological neuron
  - "Neurons are the basic signaling units of the nervous system" and "each neuron is a discrete cell whose several processes arise from its cell body"

ANN: The Biological Model Contd

- The neuron has three main regions to its structure
  - The cell body, or soma,
  - the dendrites and
  - the axon



ANN: The Biological Model Contd

- The cell body
  - is the heart of the cell, containing the nucleus and maintaining protein synthesis
  - It may have many dendrites, which branch out in a treelike structure, and receive signals from other neurons
  - It usually only has one axon which grows out from a part of the cell body called the axon hillock
  - The axon conducts electric signals generated at the axon hillock down its length
    - These electric signals are called action potentials
  - The other end of the axon may split into several branches, which end in a presynaptic terminal
  - Action potentials are the electric signals that neurons use to convey information to the brain
  - All these signals are identical

ANN: The Biological Model Contd

- Therefore, the brain determines what type of information is being received based on the path that the signal took
- The brain analyzes the patterns of signals being sent and from that information it can interpret the type of information being received
- Myelin is the fatty tissue that surrounds and insulates the axon
- Often short axons do not need this insulation
- There are uninsulated parts of the axon
  - These areas are called Nodes of Ranvier
- At these nodes, the signal travelling down the axon are regenerated
  - This ensures that the signal travelling down the axon travels fast and remains constant (i.e. very short propagation delay and no weakening of the signal)

ANN: The Biological Model Contd

- The synapse is the area of contact between two neurons
  - The neurons do not actually physically touch.
  - They are separated by the synaptic cleft, and electric signals are sent through chemical interaction
  - The neuron sending the signal is called the presynaptic cell and
  - the neuron receiving the signal is called the postsynaptic cell
- The signals are generated by the membrane potential, which is based on the differences in concentration of sodium and potassium ions inside and outside the cell membrane

ANN: The Biological Model Contd

- Neurons can be classified by their number of processes (or appendages), or by their function
  - they fall into three categories (number of processes (or appendages))
    - Unipolar neurons have a single process (dendrites and axon are located on the same stem), and are most common in invertebrates
    - In bipolar neurons, the dendrite and axon are the neuron's two separate processes. Bipolar neurons have a subclass called pseudo-bipolar neurons, which are used to send sensory information to the spinal cord
    - Finally, multipolar neurons are most common in mammals
      - Examples of these neurons are spinal motor neurons, pyramidal cells and Purkinje cells (in the cerebellum).

ANN: The Biological Model Contd

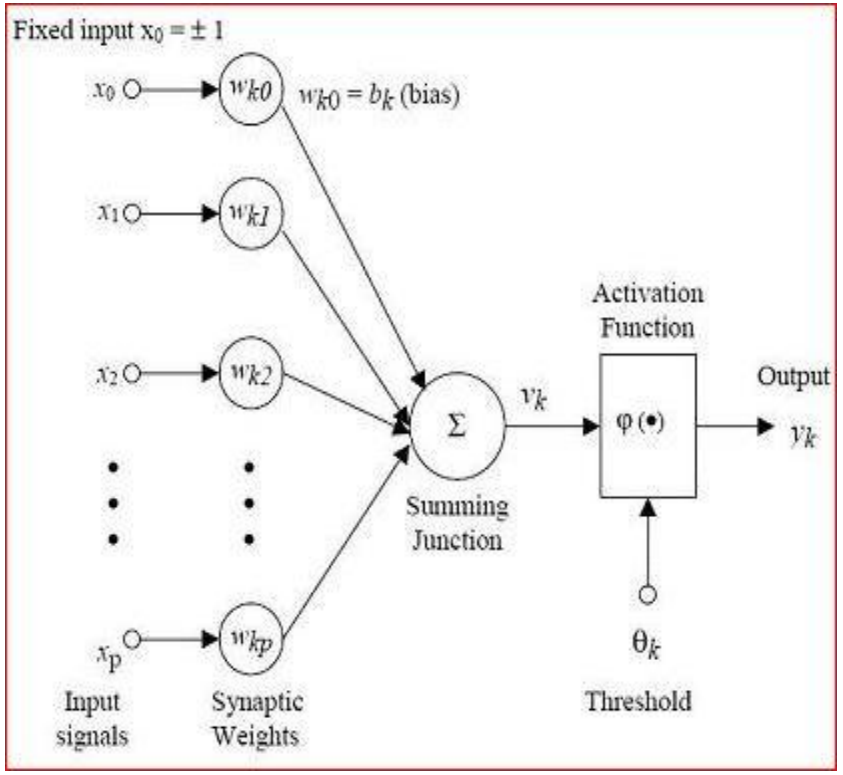
- If classified by function, neurons again fall into three separate categories
  - The first group is sensory or afferent neuron
    - which provide information for perception and motor coordination
  - The second group is called motor neuron
    - Which provides information (or instructions) to muscles and glands
  - The last group is called interneuronal, connects all other neurons and has two subclasses
    - relay or projection interneurons have long axons and connect different parts of the brain
    - local interneurons are only used in local circuits

ANN: The Mathematical Model

- When creating a functional model of the biological neuron, there are three basic components of importance
  - First, the synapses of the neuron are modeled as weights
    - The strength of the connection between an input and a neuron is noted by the value of the weight
    - Negative weight values reflect inhibitory connections, while positive values designate excitatory connections
  - The next two components model the actual activity within the neuron cell
    - An adder sums up all the inputs modified by their respective weights
      - This activity is referred to as linear combination
  - Finally, an activation function
    - controls the amplitude of the output of the neuron
    - acceptable range of output is usually between 0 and 1, or -1 and 1

ANN: The Mathematical Model Contd

- Mathematically, this process is described as:



- From this model the interval activity of the neuron can be shown to be:

$$v_k = \sum_{j=1}^{p} w_{kj} x_j$$

- The output of the neuron, y_k, would therefore be the outcome of some activation function on the value of v_k

ANN: The Mathematical Model Contd

- As mentioned previously, the activation function acts as a squashing function, such that the output of a neuron in a neural network is between certain values (usually 0 and 1, or -1 and 1)
- In general, there are three types of activation functions, denoted by Φ(.)

- First, there is the **Threshold Function** which takes on a value of 0 if the summed input is less than a certain threshold value (v), and the value 1 if the summed input is greater than or equal to the threshold value

$$\varphi(v) = \begin{cases} 1 & \text{if } v \geq 0 \\ 0 & \text{if } v < 0 \end{cases}$$

ANN: The Mathematical Model Contd

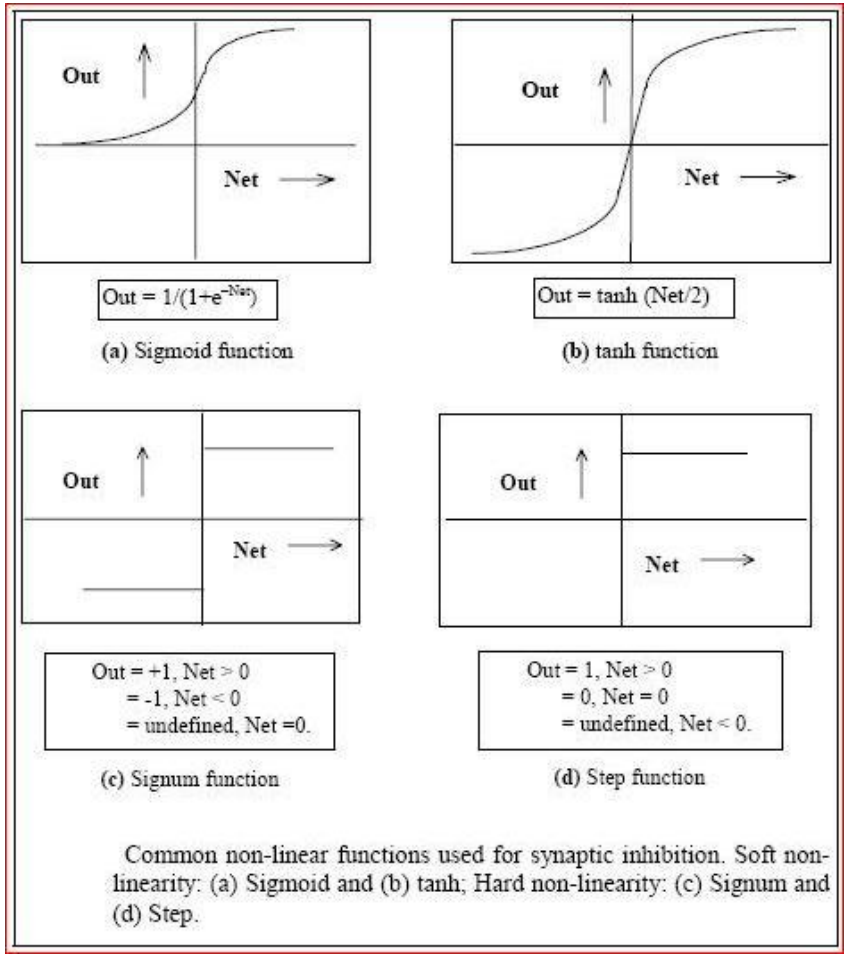
- Secondly, there is the **Piecewise-Linear function**
- This function again can take on the values of 0 or 1, but can also take on values between that depending on the amplification factor in a certain region of linear operation

$$\varphi(v)=\begin{cases}1 & v\geq\frac{1}{2}\\ v & -\frac{1}{2}>v>\frac{1}{2}\\ 0 & v\leq-\frac{1}{2}\end{cases}$$

ANN: The Mathematical Model Contd

- Thirdly, there is the **sigmoid function**
- This function can range between 0 and 1, but it is also sometimes useful to use the -1 to 1 range
- An example of the sigmoid function is the hyperbolic tangent function

$$\varphi(v) = \tanh\left(\frac{v}{2}\right) = \frac{1-\exp(-v)}{1+\exp(-v)}$$



Common non-linear functions used for synaptic inhibition. Soft non-linearity: (a) Sigmoid and (b) tanh; Hard non-linearity: (c) Signum and (d) Step.

ANN: The Mathematical Model Contd

- The ANNs which we describe are all variations on the parallel distributed processing (PDP) idea
  - The architecture of each neural network is based on very similar building blocks which perform the processing
- Lets discuss:
  - processing units
  - different neural network topologies and
  - Learning/Training strategies as a basis for an adaptive system

A Framework for Distributed Representation

- An ANN consists of a pool of simple processing units which communicate by sending signals to each other over a large number of weighted connections
- A set of major aspects of a parallel distributed model can be distinguished:
  - a set of processing units ('neurons,' 'cells')
  - a state of activation y_k for every unit, which equivalent to the output of the unit
  - connections between the units
    - Generally each connection is defined by a weight w_{jk} which determines the effect which the signal of unit j has on unit k
  - a propagation rule, which determines the effective input s_k of a unit from its external inputs
  - an activation function F_k, determines the new level of activation based on the effective input $s_k(t)$ and the current activation $y_k(t)$ (i.e., the update);
  - an external input (aka bias, offset) $ø_k$ for each unit
  - a method for information gathering (the learning rule)
  - an environment within which the system must operate, providing input signals and if necessary error signals

Processing Units

- Each unit performs a relatively simple job: receive input from neighbours or external sources and use this to compute an output signal which is propagated to other units
- Apart from this processing, a second task is the adjustment of the weights
- The system is inherently parallel in the sense that many units can carry out their computations at the same time
- Within neural systems it is useful to distinguish three types of units:
  - input units (indicated by an index i) which receive data from outside the neural network
  - output units (indicated by an index o) which send data out of the neural network, and
  - hidden units (indicated by an index h) whose input and output signals remain within the neural network
- During operation, units can be updated either synchronously or asynchronously
  - synchronous updating, all units update their activation simultaneously
  - asynchronous updating, each unit has a (usually fixed) probability of updating its activation at a time t, and usually only one unit will be able to do this at a time
  - In some cases the latter model has some advantages

Neural Network Topologies

- This section focuses on the pattern of connections between the units and the propagation of data
- As for this pattern of connections, the main distinction we can make is between:
  - **Feed-forward neural networks**
    - where the data flow from input to output units is strictly feed forward
    - The data processing can extend over multiple (layers of) units, but no feedback connections are present
      - that is, connections extending from outputs of units to inputs of units in the same layer or previous layers

Neural Network Topologies Contd

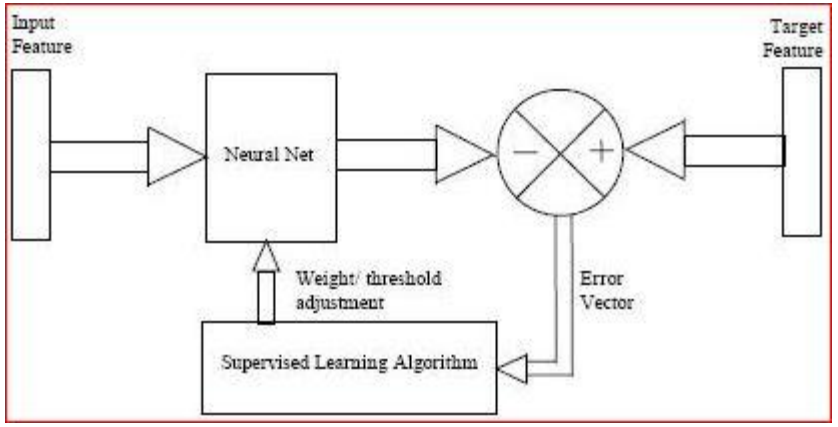
- **Recurrent neural networks**
  - contains feedback connections, contrary to feed-forward networks
  - the dynamical properties of the network are important
  - In some cases, the activation values of the units undergo a relaxation process such that the neural network will evolve to a stable state in which these activations do not change anymore
  - In other applications, the change of the activation values of the output neurons are significant, such that the dynamical behaviour constitutes the output of the neural network

Training of Artificial Neural Networks

- A **neural network** has to be configured such that the application of a set of inputs produces (either 'direct' or via a relaxation process) the desired set of outputs
- Various methods to set the strengths of the connections exist
  - One way is to set the weights explicitly, using a priori knowledge
  - Another way is to **'train' the neural network** by feeding it teaching patterns and letting it change its weights according to some learning rule
- We can categorise the learning situations in three distinct sorts. These are:
  - **Supervised learning** or Associative learning
  - **Unsupervised learning** or Self-organization
  - **Reinforcement Learning**

Training of Artificial Neural Networks: Supervised Learning

- **Supervised learning** or Associative learning in which the network is trained by providing it with input and matching output patterns
  - These input-output pairs can be provided by an external teacher, or by the system which contains the neural network (self-supervised)



Training of Artificial Neural Networks: Unsupervised Learning

- **Unsupervised learning** or Self-organisation in which an (output) unit is trained to respond to clusters of pattern within the input
- In this paradigm the system is supposed to discover statistically salient features of the input population
- Unlike the supervised learning paradigm, there is no apriori set of categories into which the patterns are to be classified; rather the system must develop its own representation of the input stimuli

Training of Artificial Neural Networks: Reinforcement Learning

- **Reinforcement Learning**
  - This type of learning may be considered as an intermediate form of the above two types of learning
  - Here the learning machine does some action on the environment and gets a feedback response from the environment
  - The learning system grades its action good (rewarding) or bad (punishable) based on the environmental response and accordingly adjusts its parameters
  - Generally, parameter adjustment is continued until an equilibrium state occurs, following which there will be no more changes in its parameters

Thank you